

UNIT 02:

Chapter 01: [Design Domain]

Lesson 01: [Lets make it fit]

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson, students will be able to:

Understand the importance of precise measurements in 3D modeling.

Explain how components rotate around an axis and why proper fitting is necessary.

Use Tinkercad to modify a pulley and create a precise hole for a motor shaft.

Demonstrate their ability to apply engineering concepts in digital design.

Challenge questions of the lesson:

If the pulley hole is too large or too small for the motor shaft, how can you modify your Tinkercad design to ensure a perfect fit? What factors should you consider when resizing the hole?"

Expected Answer:

If the hole is too **small**, students should **increase** its diameter slightly, ensuring it matches the **exact measurement** of the motor shaft.

If the hole is too **large**, students can **decrease** its size or consider adding a **locking mechanism** such as a set screw or a keyway in Tinkercad.

ESL Relation:

Vocabulary Support: Key terms with visuals (pulley, motor shaft, precision, axis).

Language Scaffolding: Sentence starters, word wall, graphic organizers.

Visual & Hands-On Learning: Use Tinkercad, real-world demonstrations, peer collaboration.

Speaking & Listening Practice: Think-Pair-Share, verbal explanations before writing.

Reading & Writing Support: Simplified instructions, labeled diagrams, step-by-step guidance.

Answers of exercises:

Let's Think:

Question: Look at the part inside the circle; it's called the motor shaft. What shape is it?

Answer: The motor shaft is cylindrical in shape.

Explore - Do It:

Question: Create a new hole in your pulley to fit the shaft of the motor.

Answer: Students will create a precise hole in Tinkercad by measuring the diameter of the motor shaft and adjusting the hole size in their pulley design.

Showcase:

Question: Change the pulley and add the motor to your machine.

Answer: After designing the hole in Tinkercad, students will adjust the pulley to fit the motor shaft properly and integrate it into their machine.

Lesson Plan

1: ENGAGE (15 minutes) – Spark Curiosity

From the Book: "Let's Think" & Stepping Stone

Goal: Help students understand the importance of precision in engineering and introduce the problem-solving process.

- Introduce a real-world problem: **Leila notices the motor doesn't fit.**
- Ask students why engineers need to be precise in their designs.
- Show a pulley in **Tinkercad** and compare it to a real pulley.
- Play the **QR code video** from the book to introduce how to create a motor shaft hole.

Teacher's Role:

- Present a **Tinkercad pulley design** and ask students to predict what happens if the hole is **too small or too big.**
- Guide a discussion on **why precision is essential** in engineering.
- Relate the concept to real-world examples like misfitting mechanical parts.

Expected from Students:

- Share ideas on **why precision matters.**
- Discuss **real-life examples** of objects that don't fit properly.

Key Questions:

- What challenges arise when a part does not fit properly?
 - How do engineers ensure precision in designs?
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2: EXPLORE (20 minutes) – Hands-On Investigation

From the Book: "Explore" Section

Goal: Allow students to experiment with creating precise holes in a pulley using **Tinkercad.**

- Students **open Tinkercad** and import or create a pulley model.
- They **check the hole size** by measuring the motor shaft.
- Experiment with **different hole sizes** using **Tinkercad** tools.

Teacher's Role:

- Provide students with the **correct motor shaft dimensions.**
- Demonstrate how to **adjust a hole size** in **Tinkercad.**
- Observe students' **trial-and-error process** and provide **guidance.**

Expected from Students:

- Try **different hole sizes** in their design.
- Measure and **compare** their digital design with real-world sizes.

Key Questions:

- How does changing the hole size affect the pulley's function?
 - What happens if the hole is too loose or too tight?
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3: EXPLAIN (20 minutes) – Concept Understanding

From the Book: "Stepping Stone" - Explanation

Goal: Deepen students' understanding of **measurement, tolerances, and rotational movement**.

- Discuss how a **pulley rotates** around a motor shaft.
- Introduce **measurement tools in Tinkercad**.
- Explain the **importance of tolerances** in design (**not too tight, not too loose**).

Teacher's Role:

- Show how to use **Tinkercad's ruler and alignment tools** for precision.
- Ask students to **justify their design choices** (e.g., Why did you choose this hole size?).
- Connect the learning to **real-world manufacturing** and **engineering design standards**.

Expected from Students:

- Explain their **design decisions**.
- Discuss the **impact of precise measurements**.

Key Questions:

- Why is it important for engineers to use measurement tools?
 - How do tolerances affect the functionality of a product?
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4: ELABORATE (25 minutes) – Apply Learning

From the Book: "Showcase" - Modify & Test

Goal: Apply the learned concepts to **refine and test a pulley design** in Tinkercad.

- Students **finalize their pulley design** in Tinkercad.
- Run a **Tinkercad simulation** to see how the pulley fits.
- Check their **dimensions** against the given **motor shaft size**.

Teacher's Role:

- Provide **feedback** on student designs.
- Ask students **how they verified the fit**.
- Support students in making **adjustments and refinements**.

Expected from Students:

- Modify and **refine their designs**.
- Ensure their **hole fits correctly** in the **3D model**.

Key Questions:

- What steps did you take to ensure the pulley hole fits perfectly?
- How does the Tinkercad simulation help verify your design?

5: EVALUATE (10 minutes) – Assess Learning

From the Book: "Now I Can..." - Self-Assessment

Goal: Assess students' understanding of **precision and design modifications** in Tinkercad.

- Students **reflect on what they learned**.
- Answer **challenge questions**:
 - Why do we need to measure before designing the hole?
 - How does Tinkercad help engineers create precise designs?

Teacher's Role:

- Facilitate a **discussion** based on student reflections.
- Check **students' final pulley designs**.
- Provide **constructive feedback** for further improvement.

Expected from Students:

- Answer **challenge questions** with **reasoning**.
- Reflect on their **design improvements**.

Key Questions:

- How did your final design differ from your initial attempt?
 - What challenges did you face, and how did you solve them?
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❖ **Relation to Standards:**

- NGSS (Next Generation Science Standards)
- CCSS (Common Core State Standards - Math)
- ISTE (International Society for Technology in Education)
- ELPS (English Language Proficiency)

UNIT 02:

Chapter 02: [Mechanics Domain]

Lesson 01: [Final Step]

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson, students will be able to:

Define what a magnet is and describe its basic properties.

Observe and understand magnetic forces by testing different materials.

Identify magnetic and non-magnetic materials through hands-on experiments.

Apply their knowledge by incorporating magnets into a sorting machine.

Think critically about why magnets attract certain objects.

Answers of exercises:

Focus Section:

Which of these is NOT magnetic?

Correct Answer: B)

Wooden block

Which of these is magnetic?

Correct Answer: D) Nails

Explore Section (Observation Table)

Students were asked to test objects and mark whether they are **magnetic** (✓) or **non-magnetic** (✗)

Challenge questions of the lesson:

Why do you think magnets can pull things to them?

Answer: "Magnets create an invisible force called a **magnetic field**. This field attracts certain materials, like iron, nickel, and cobalt, but does not affect wood, plastic, or glass."

Why do some objects get attracted to a magnet while others don't?

Answer: "Only materials that contain certain metals, like **iron, nickel, or cobalt**, are attracted to magnets. Other materials, like **wood, plastic, and paper**, do not have the right properties to respond to a magnetic field."

ESL Relation:

Visual aids – Diagrams, videos, real-life objects.

Hands-on learning – Engaging physically with materials.

Simple language – Breaking down complex ideas.

Pre-teach key vocabulary – Using word walls and flashcards.

Lesson Plan:

1. Engage (15 minutes)

Book Connection: Stepping Stone Section (Story of Magnus)

Goal: Spark curiosity through storytelling and demonstrations.

Teacher's Role:

- **Tell the story of Magnus from the book:**
 - “There was once a shepherd named Magnus who lived in Magnesia, Greece. One day, Magnus was herding his sheep through the mountains. Suddenly, he noticed that the tip of his stick and the nails in his sandals got stuck to a rock.”
- **Ask students:**
 - What do you think is happening in this story?
 - Have you ever seen something like this before?
- **Demonstration:**
 - Take a **magnet** and show how it attracts a **paperclip** but **not a wooden block**.
 - Encourage predictions about which objects might be **magnetic**.

Expected Student Engagement:

- Listen and engage in the story discussion.
- Make guesses about what is happening in the story.
- Offer predictions about magnetic objects.

Key Questions to Ask:

- Why do you think some objects are attracted to the magnet and others are not?
 - What do you think happened to Magnus' stick and sandals?
 - Where have you seen magnets used in real life?
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2. Explore (25 minutes)

Book Connection: *Explore Section (Testing Materials)*

Goal: Let students discover magnetic and non-magnetic objects through hands-on activities.

Teacher's Role:

- **Distribute materials:**
 - Provide each group with a **magnet** and a variety of **small objects**.
- **Guide hands-on investigation:**
 - Ask students to test each object by bringing it close to the magnet.
 - Have them **record their observations** in the **table from the book's "Explore" section** (✓ for magnetic, ✗ for non-magnetic).
- **Encourage discussion:**
 - Ask groups to **compare results** and find patterns.

Expected Student Engagement:

- Work in small groups.

- Test objects and record observations.
- Discuss findings and compare results with classmates.

Key Questions to Ask:

- What do all the magnetic objects have in common?
 - Why do some materials not stick to the magnet?
 - What happens if we try to put two magnets together?
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3. Explain (20 minutes)

Book Connection: Focus Section (Multiple-Choice Questions & Definitions)

Goal: Help students connect their observations to scientific concepts.

Teacher's Role:

- **Facilitate discussion:**
 - Ask groups to **share their results**.
 - Highlight key findings: “All the metal objects were magnetic, except for aluminum foil. Why might that be?”
- **Introduce scientific vocabulary:**
 - **Magnetic field** – the invisible force around a magnet.
 - **Attraction & Repulsion** – magnets can both pull and push.
 - **Ferromagnetic materials** – iron, nickel, and cobalt.
- **Use book exercises:**
 - Guide students through the "**Focus**" section in the book.
 - Question 1: Which of these is NOT magnetic?
 - Question 2: Which of these is magnetic?

Expected Student Engagement:

- Share findings.
- Take notes on key concepts.
- Answer book questions.

Key Questions to Ask:

- Why do magnets only attract certain metals?
 - How can we use magnets in everyday life?
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4. Elaborate (20 minutes)

Book Connection: Practice Section (Building a Sorting Machine)

Goal: Apply knowledge to a real-world scenario.

Teacher's Role:

- **Introduce the challenge:**
 - "Imagine you are engineers designing a machine to sort objects using magnets."
 - Show the **sorting machine model** (from the book's "Practice" section).
- **Guide students:**
 - Ask them where they would place **magnets** to make the machine work best.
 - Encourage teamwork and **problem-solving**.
- **Facilitate testing:**
 - Allow students to test their sorting machines with different objects.
 - Guide them in making improvements.

Expected Student Engagement:

- Work in teams to modify the sorting machine.
- Test and improve their design.

Key Questions to Ask:

- Why is the placement of the magnet important?
 - How could this type of machine be useful in a recycling plant?
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5. Evaluate (10 minutes)

Book Connection: Let's Think Section (Challenge Question & Reflection)

Goal: Assess student understanding through discussion and final questions.

Teacher's Role:

- **Challenge Question (from the book):**
 - Ask: Why do you think magnets can pull things to them?
 - Expected Answer: "Magnets create an invisible force field that attracts certain materials, like iron, nickel, and cobalt."
- **Review results:**
 - Go over the observation table and sorting machine designs.
- **Self-reflection:**
 - Ask students to write or share what they learned today.

Expected Student Engagement:

- Answer the final challenge question.
- Reflect on their learning experience.

Key Questions to Ask:

- Can you think of a place where magnets are used in real life?
- What is one thing you learned today that surprised you?

❖ Relation to Standards:

- Science Standards (NGSS - Next Generation Science Standards)
- Common Core State Standards (CCSS - English & Math Integration)
- English Language Proficiency (ELP) Standards

Theme 1

[Sorting magnetic and non-magnetic materials]

UNIT 03:

Chapter 01: [Programming]

- **Lesson 01: Learn and play**
- **Time frame: 80 mins.**

Lesson objectives:

By the end of this lesson, students will be able to:

1. **Add and customize sprites** in a programming environment.
2. **Use motion blocks** to make sprites move, turn, and interact.
3. **Define an algorithm** as a step-by-step sequence of instructions.
4. **Understand variables** as containers that store changing values (e.g., score).

Challenges questions:

- "What blocks would make the bin move left when the left arrow is pressed?"
- "How could you make items fall faster over time?"
(Hint: Use a speed variable that increases!)
- "Write the algorithm for what happens when the bin catches paper:"
- "How could you add a 'bonus recyclable' (like aluminum cans) that gives extra points?"

Sel relations:

1. SELF-AWARENESS (Identifying Emotions & Growth Mindset)

- **Example:** After debugging failed code, ask:
"How did you feel when your game didn't work? What helped you keep trying?"
- **Strategy:** Use an "Emotion Thermometer" poster (☐ frustrated → ☐ proud) for students to point to their feelings during challenges.

2. SELF-MANAGEMENT (Perseverance & Problem-Solving)

- **Example:** Normalize mistakes:
"Game developers debug 100 times before it works! Let's celebrate 'glitches' as learning steps."
- **Strategy:** Provide a "Fix-It Kit" (printed common bug solutions) to promote independence.

3. SOCIAL AWARENESS (Collaboration & Empathy)

- **Example:** Pair programming roles:
 - **Driver:** Controls the keyboard.
 - **Navigator:** Gives instructions (*"Try a 'repeat until' block here!"*).
- **Strategy:** Assign "Help Tickets"—students must ask peers before the teacher.

4. RELATIONSHIP SKILLS (Communication & Teamwork)

- **Example:** Group reflection:
"How did your partner's idea improve your game?"
- **Strategy:** "Kind-Question-Suggestion" feedback during game demos:

- "I like how you added sound effects!"
- "What if the bin changed color when it lost a life?"

Engage:

□□ TEACHER ROLE

1. Engage (Hook & Predictions)

- **Demo the Game:** Show a working version. Ask:
"What happens when the bin touches plastic vs. banana? Why?"
- **Introduce Variables:**
"This 'score' number changes when you catch recyclables—it's a variable!"

2. Facilitate (Coding Support)

- **Guide Algorithm Design:**
"What steps make items fall? How can we detect collisions?"
- **Debugging Prompts:**
"Your banana isn't disappearing. Check if you're using 'delete this clone'!"

3. Support SEL

- **Model Persistence:**
"My game glitched at first—I fixed it by testing small parts!"
- **Encourage Peer Help:**
"Ask a neighbour before me! Use phrases like 'Could we try...?'"

□ STUDENT ROLE

1. Explore & Build

- **Test Movement:**
"Make the bin move left/right with arrow keys."
- **Experiment with Variables:**
"Change 'score' to +2 for plastic, -1 for banana."

2. Collaborate & Debug

- **Team Roles:**
 - **Algorithm Designer** (plans steps).
 - **Variable Manager** (tracks score/lives).
- **Troubleshoot:**
"Why does the score reset? Check where 'set score to 0' is!"

3. Reflect & Share

- **Explain Code:**
"Our algorithm: 1) Items fall, 2) Bin checks collisions, 3) Score updates."
- **Celebrate Teamwork:**
"We combined ideas to make the game harder!"



let's Think



Arrange the steps to help me understand how can we make a game?



Make the code.



Play the game.



Make the design.

Explore



Choose what your game will look like and what will be in it by choosing your sprites and background.



Watch it..

This video will help you understand more about how to make the game.



Scan Here!

<https://www.youtube.com/watch?v=QMZtfeGT75c>



In this part this the link of game:

https://drive.google.com/drive/folders/1lBGey2c_5crmajC6q8LUT-zkJh3Wlpu?usp=drive_link

Assessment

Focus



In mBlock, which category would you find the block to create a variable for counting?

- A) Looks.
- B) Events.
- C) Variables.
- D) Sensing.